

# Analogue Circuit Design

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## Engineering Technology

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# Analogue Circuit Design (A04589)

**Short Title:** Analogue Circuit Design  
**Faculty** Science and Computing  
**Department** Engineering Technology  
**Cluster** Engineering  
**Author** CLAIRE.FITZPATRICK  
**Credits** 5

**Level:** Advanced

## Description of Module / Aims

This module covers analogue circuit design techniques and sensor and transducer signal conditioning circuits. 1. Introduction to modern analogue sensors, transducers and their characteristics 2. Analogue circuit design techniques with emphasis on transducer signal conditioning, bridge circuits, etc. 3. Analogue-to-Digital and Digital-to-Analogue converter fundamentals and architectures 4. Basic introduction to mixed signal system-on-chip and IoT signal processing hardware platforms.

## Programmes

|           |                                                    |           |
|-----------|----------------------------------------------------|-----------|
|           | BEng (Hons) in Electronic Engineering (WD_EEELC_B) | 3 / 5 / E |
| COMP-0565 | BSc (Hons) in Applied Computing (WD_KCOMP_B)       | 4 / 7 / E |
| ANCD-0001 | BSc (Hons) in Applied Electronics (WD_EELEC_B)     | 1 / 1 / M |

## Indicative Content

- Introduction to analogue sensors and transducers (temperature, humidity, strain, pressure, magnetic, output transducers, relays, etc)
- Bridge sensor configurations (single, dual, quad, linear pot, active, centre tapped)
- Analogue circuit design and signal conditioning for sensor applications (bridge excitation, ratiometric operation, operational amplifier types and specifications, instrumentation amplifiers, isolation amplifiers, cmv, cmr)
- Introduction to A/D and D/A converters (fundamentals, types, architectures, operation, characteristics, errors, signal sampling)
- Introduction to mixed signal system-on-chip and IoT hardware (e.g. PSoC-4 BLE, Arduino, Raspberry Pi, mBed, Waspote, etc)
- Circuit design and simulation skills using analogue and mixed mode (AMS) simulation CAD tools
- Design, build and test of a sensor system through the mini-project assessment element of the module

## Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Compare and contrast the various types of sensors and bridge sensor configurations and analyse their performance characteristics.
2. Solve a variety of sensor design problems by applying appropriate analogue signal conditioning solutions.
3. Compare and contrast the operation and characteristics of different data converter architectures and evaluate key performance parameters.
4. Appraise mixed-signal System-on-Chip devices (e.g. PSoC) and IoT sensor technology hardware.
5. Design and analyse various analogue and mixed signal circuits using Spice simulation software.
6. Design, simulate, build, & test a basic sensor system using a mixed signal hardware platform.

## Learning and Teaching Methods

- Lectures
- Discussion and worked examples
- CAD laboratory design and simulation sessions
- Practical laboratory-based mini-project. Hardware design, build and test of a basic sensor system using a mixed signal hardware platform.
- Q & A Sessions (class or web)

## Assessment Methods

|                            | Weighing | Outcomes Assessed |
|----------------------------|----------|-------------------|
| Continuous Assessment      | 100%     |                   |
| <i>In-Class Assessment</i> | 25%      | 1,2               |
| <i>In-Class Assessment</i> | 25%      | 3,4               |
| <i>Lab Report</i>          | 5%       | 1,5               |
| <i>Lab Report</i>          | 5%       | 2,5               |
| <i>Project</i>             | 40%      | 6                 |

## Assessment Criteria

### Fail

(<40%): Unable to interpret and specify key concepts of sensor technology, signal conditioning, or programmable sensor hardware.

### Notes:

- Pass (40% - 49%): Able to interpret and specify the key concepts of sensor technology, signal conditioning, or programmable sensor hardware.
- Second Class Honours - Grade 2 (50% - 59%): Ability to specify key concepts of sensor technology, signal conditioning circuitry and programmable sensor hardware by demonstrating an ability to locate and utilise this knowledge within a broader mixed signal design environment.
- Second Class Honours - Grade 1 (60% - 69%): Be able to solve sensor design problems, analyse sensor technology, signal conditioning circuitry, or programmable hardware and demonstrate an ability to deploy this knowledge within a broader mixed signal design environment.
- First Class Honours (70% - 100%): All the above to an excellent level. Be able to analyse and design solutions to a high standard for a range of both complex and unforeseen problems through the use of appropriate skills and tools. An ability to synthesise solutions to problems should be evident here.

## Learning Modes

| Learning Mode        | F/T Hours | P/T Hours |
|----------------------|-----------|-----------|
| Lecture              | 24        |           |
| Practical            | 24        |           |
| Independent Learning | 87        |           |

## Essential Material

- Book. Horowitz, P. and W. Hill. *The Art of Electronics*. 3rd ed. .: Cambridge University Press, 2015.

- Components, Electronic. \_Sensors, op-amps, data converters, passive components, etc.\_. ..
- Hardware, Electronic. \_Sockets, connectors, terminals, prototyping boards, etc.\_. ..
- Instrumentation, Test. \_Scopes, Signal generators, Logic labs, Mini-fridges, Precision thermometer\_. ..
- Software, CAD. \_PSoC-4 BLE Pioneer Development Kit or similar\_. ..
- Software, CAD. \_Schematic Capture and Spice circuit simulation software\_. ..

## Supplementary Material

- Fraden, J. \_Handbook of Modern Sensors: Physics, Design and Applications\_. 4th ed. New York: Springer, 2010.
- Jung, W. \_Op Amp Applications Handbook\_. : Newnes, 2004.
- Kester, W. \_Data Conversion Handbook by Analog Devices Staff\_. 3rd ed. : Newnes, 2004.

## Requested Resources

- Lecture Room: Loose Seated
- ENGINEERING LAB: Electronics